Coding is the process of making sense of qualitative primary data by assigning meaningful labels to portions of the data so that patterns, categories, and themes can be identified. Denny and Clark explain that coding involves assigning a shorthand designation to parts of the data so researchers can retrieve and compare specific pieces of information (Denny and Clark 92). Their discussion also shows that coding is not simply a matter of labeling information mechanically; it is an interpretive process that connects the data to the research question. Downs and Wardle similarly emphasize that primary research should involve generating knowledge from firsthand evidence rather than simply collecting facts from other researchers. They describe students using interviews, surveys, observations, and discourse analysis and stress that careful observation and empirical data gathering can strengthen a researcher's authority (Downs and Wardle 563). Together, these ideas show me that coding is one part of a larger process of turning firsthand observations or responses into evidence-based knowledge.

I would first make sure my research question is clear and can guide my analysis. Then I would organize my data in one place and remove any names or other identifying information before I begin coding. Denny and Clark explain that researchers should anonymize primary data through pseudonyms, numbers, or codes because participants' identities can affect confidentiality and can also introduce researcher bias into the analysis (Denny and Clark 91). After preparing the data, I would read through all of it once without trying to force it into categories. My goal would be to understand the overall meaning and notice repeated ideas, surprising responses, or patterns. Denny and Clark describe this as getting a general sense of the data before moving into detailed coding (92). I would make brief notes during this first reading and mark recurring ideas with simple labels.

Next, I would conduct a more detailed round of coding. Rather than deciding in advance exactly what every response should mean, I would allow codes to emerge from what participants actually say, especially if I am using open coding. For example, if my research question concerned students' experiences with online learning, responses mentioning difficulty concentrating might receive a code such as "distraction," while responses describing communication with classmates might receive "connection." Denny and Clark demonstrate this process by assigning descriptive words and phrases to excerpts based on what the participants' comments actually communicate (Denny and Clark 94–95). I would then compare similar codes and combine them into broader categories or themes. I would also keep track of why I assigned particular codes so that my interpretation can be explained rather than appearing arbitrary. If possible, I would have another person independently review some of the data or coding. Denny and Clark explain that comparing coding decisions can provide another perspective, reduce researcher bias, and help establish qualitative validity (96).

After identifying the strongest categories and themes, I would interpret what those patterns mean in relation to my research question. I would look for ideas that appear

repeatedly rather than building my conclusions around one unusual response. I would also pay attention to contradictory or unexpected evidence instead of ignoring it. Denny and Clark warn against “cherry picking,” or making claims from data that are too thin or not representative of the larger dataset (97). This means that my final report should distinguish between what my data actually shows and what I might reasonably infer from it. Downs and Wardle’s emphasis on primary research as a way of generating knowledge is important here because the goal is not just to repeat what other people have said; it is to use my own evidence to contribute something to the conversation (563).

Ethics are especially important when interpreting primary data because the data often comes from real people. I would protect participants’ privacy by removing identifying information and using pseudonyms or participant numbers. I would also avoid changing participants’ words to make them fit my argument. Ethical analysis requires representing the data honestly, including evidence that complicates my interpretation. Denny and Clark explain that researchers should make only claims they can support and should avoid exaggerating or misleading readers (Denny and Clark 97). For example, I should report the size and context of my sample instead of presenting a small percentage as though it represents a much larger population. I should also avoid selecting only the most interesting quotations while ignoring responses that point in another direction. I would keep a clear record of how I analyzed the data and reached my conclusions to help readers understand my findings and show that I treated the participants and their data with care.